Project Description

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Setup of the Original Model as in @Agu2023

We implement the dynamic discrete-choice demand model of @Agu2023. A consumer i each period t chooses $y_{it} \in \{0, 1, \dots, J\}$ where $y_{it} = 0$ is "no purchase" and $y_{it} = j > 0$ is buying brand j. The endogenous state $x_{it} = (\ell_{it}, d_{it})$ records the last brand purchased and the duration since the last purchase. Prices have a persistent component z_{it} and a transitory (promotion) component e_{it}. With i.i.d. Type-I Extreme Value shocks the per-period payoff absent the shock is

$$u_{\alpha_i}(y, x, p) = \begin{cases} \alpha_i(\ell) + \gamma h(\mu_i) - \beta^{dep}(\ell)\, d & y = 0, \\ \alpha_i(j) + \gamma h(\mu_i - p(j)) - \beta^{sc}(\ell, j) & y = j > 0. \end{cases}$$

The block below sets up the primitives, solves the consumer's Bellman equation by VFI (used only for simulating data), simulates a panel, and estimates $\theta = (\gamma, \beta^{sc}, \beta^{dep})$ by the sufficient-statistic Conditional Maximum Likelihood (CML) estimator described in Sections 3–4 of the paper.

Table 1: Simulation and CML estimation summary.

Statistic	Value
Consumers	25000
Periods	52
Brands	2
Observed purchases	677750 of 1300000
Purchase rate	52.1%
Sufficient-statistic cells with $e_3 = e_4$	14744
Price-contrast std. dev.	0.095
Nonzero price-contrast share	42.5%
Alternating-history share	26.1%
Depreciation spell cells	3449
Depreciation duration cap d^*	2
Minimum T for depreciation CML	5
Constant-promotion spell restriction	True
Alpha FE sample std. dev.	0.249, 0.453
Optimizer converged?	Yes
Optimizer iterations	11

Table 2: CML parameter estimates.

Parameter	True	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%	Bias
γ	1.0350	0.8146	0.1989	4.0962	0.0000	0.4248	1.2044	-0.2204
$\tilde{\beta}_{1,2}$	1.0500	1.0398	0.0188	55.4026	0.0000	1.0030	1.0766	-0.0102

Table 3: CML depreciation estimates from no-purchase spell histories.

Parameter	True	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%	Bias
β_1^{dep}	0.1000	0.0335	0.0438	0.7657	0.4438	-0.0523	0.1193	-0.0665
β_2^{dep}	0.1300	0.1266	0.0543	2.3304	0.0198	0.0201	0.2331	-0.0034

Table 4: General CML fit and test statistics.

Statistic	Value
Log-likelihood	-8452.796
Null log-likelihood	-10219.762
Pseudo R^2	0.173
AIC	16909.593
BIC	16924.790
LR test statistic	3533.931
LR test df	2
LR test p-value	0.000

Note: The first CML table uses purchase histories (k, j, k, j) and (k, k, j, j) to estimate γ and $\tilde{\beta}_{k,j}$. The depreciation table uses the no-purchase spell histories from Section 3.3, imposing the simulated duration cap $d^* = 2$ as the point where the depreciation state is flat.